Ashin Murugan

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SUMMARY

Integrated M.Tech Cyber Security student with hands-on experience in cloud security (AWS, GCP), vulnerability assessments, and security automation using Python. Completed a internship with the High Court of Kerala identifying 5+ security findings of varying severity. Worked on applying machine learning to security problems, including anomaly detection and network traffic analysis, with a published research paper on using GANs for phishing simulation. Particularly interested in growing further in AI-driven security and using AI to strengthen detection, defense, and compliance systems.

EDUCATION

Vellore Institute of Technology — **CGPA: 8.74 / 10**
Integrated M.Tech – Computer Science and Engineering (Cyber Security) — Oct 2022 – Present

TECHNICAL SKILLS

- **Application Security:** Web Security, OWASP Top 10, Secure Code Review, VAPT, Threat Modeling, Security Controls Analysis
- **Security Tools:** Burp Suite, Nmap, Wireshark, Metasploit, OWASP ZAP, Wazuh SIEM, Zabbix, MS Sysinternals, Linux
- **Cloud & Platforms:** AWS (IAM, EC2, S3, VPC, Lambda), Cloud Security Architecture, SIEM Integration, Zero Trust
- **Scripting & Development:** Python, FastAPI, React, MySQL, Git
- **Networking & Forensics:** TCP/IP, Packet Analysis, Network Defense, SOC Operations, Incident Response, MITRE ATT&CK, Cryptographic Protocols, Digital Forensics, Log Analysis

EXPERIENCE

High Court of Kerala — *Ernakulam, Kerala, IN*
Cyber Security Intern — Oct 2025 – Dec 2025

- Conducted security assessments across 47+ endpoints in a regulated environment, remediating 5+ vulnerabilities while ensuring strict adherence to organizational compliance and security mandates.
- Synthesized technical findings into strategic reports, bridging raw vulnerability data with IT management goals to drastically improve long-term architectural defense and system resilience.
- Engineered Python automation scripts to streamline log collection, reducing review time by 40% and engineered a Unified Ops Dashboard for real-time visibility.

PROJECTS

PhantomFlow – ML-Based Covert Channel Detection — GitHub
Python, LSTM, XGBoost, Kafka, FastAPI, PostgreSQL — Mar 2026 – May 2026

- Engineered an ML ensemble pipeline (LSTM, XGBoost, IsolationForest) to detect C2 beaconing, DNS tunneling, and data exfiltration in encrypted traffic (TLS 1.3, QUIC, DoH) without decryption.
- Conducted cross-dataset validation across CICIDS and CTU-13, proving that 6 invariant features match the performance of 50-feature models for cross-family C2 detection.
- Designed an async Kafka-based detection pipeline with SHAP explainability and MITRE ATT&CK mapping, incorporating concept drift detection and active learning for analyst review.

AgentWall – Intelligent AI Security Layer — GitHub
Python, FastAPI, React, DuckDB, psutil — Apr 2026 – May 2026

- Engineered a security layer using heuristic process monitoring to correlate OS-level file modifications with autonomous AI agent processes (Cursor, Aider) in real-time.
- Implemented an asynchronous telemetry pipeline with DuckDB and HMAC-chained logging, handling 100+ critical security events per second with zero latency impact.
- Developed a forensic SOC dashboard visualizing MITRE ATT&CK technique mapping with step-by-step session replay for complex LLM-based interactions.

Unified Ops Security Monitoring Dashboard — GitHub
Python, FastAPI, Wazuh SIEM, Zabbix — Oct 2025 – Dec 2025

- Architected a centralized SOC dashboard integrating Wazuh SIEM and Zabbix across 13+ distributed servers, reducing alert latency to under 15 seconds for suspicious activity.
- Configured Prometheus metrics scraping for cloud workloads and developed custom Grafana visualizations to enable proactive, real-time incident detection and reporting.

ACHIEVEMENTS

- **Research Publication:** Co-authored Automated Security Surveillance Using Computer Vision in Public Spaces (ICSTSDG 2024).
- **Amazon ML Challenge 2025:** Secured 192nd global rank by building a multi-modal ML pipeline for optimal product price prediction using provided datasets.
- **Research Publication:** Published CT-GAN: A Conditional Transformer GAN for generating realistic phishing emails which can be used in employee training and threat intelligence (ICICNS 2026).

CERTIFICATIONS

- **Amazon Web Services:** AWS Certified Cloud Practitioner
- **Google Cloud:** Google Cloud Cybersecurity Certificate *(Beginner)*
- **Cisco Certified:** Ethical Hacker, Network Defense **Google:** The Bits and Bytes of Computer Networking